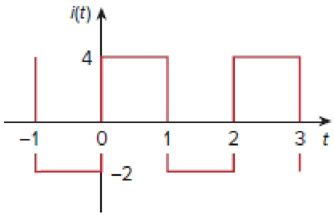
Chapter 17, Problem 47P

Problem

The periodic current waveform in Fig. 17.79 is applied across a 2-kΩ resistor. Find the percentage of the total average power dissipation caused by the dc component.

Figure 17.79:



Step-by-step solution

Step 1 of 4

Consider the following periodic current waveform:

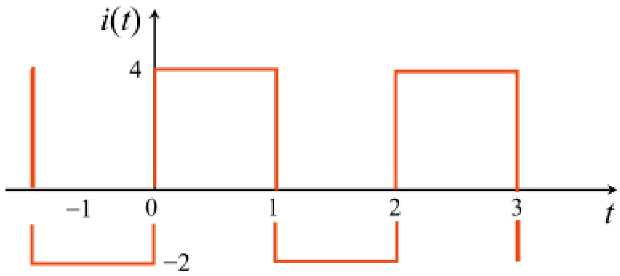


Figure 1

Step 2 of 4

The time period of Figure 1 is 2. Therefore,

$$\begin{aligned}\omega_0 &= \frac{2\pi}{T}\\ &= \frac{2\pi}{2}\\ &= \pi\end{aligned}$$

Calculate the dc component or average value a_0.

$$\begin{aligned}a_0 &= \frac{1}{T}\int_0^T i(t)dt\\ &= \frac{1}{2}\int_0^1 4dt + \frac{1}{2}\int_1^2 (-2)dt\\ &= 2t\big|_0^1 - t\big|_1^2\end{aligned}$$

$$\begin{aligned}&= 2[1-0]-[2-1]\\ &= 1\end{aligned}$$

Step 3 of 4

Calculate the power through the $2\text{ k}\Omega$ resistor.

$$\begin{aligned}
P &= i_{\text{rms}}^2 R \\
&= R\left(\frac{1}{T}\int_0^T i^2(t)\,dt\right) \\
&= \frac{R}{2}\int_0^1 (4)^2\,dt + \frac{R}{2}\int_1^2 (-2)^2\,dt \\
&= 8R + 2R \\
&= 10R
\end{aligned}$$

Step 4 of 4

Determine the average power dissipation caused by the dc component.

$$\begin{aligned}
P_0 &= a_0^2 R \\
&= (1)^2 R \quad (\text{since } a_0 = 1) \\
&= R \\
&= \frac{P}{10} \quad (\text{since } P = 10R) \\
&= 10\% \text{ of } P
\end{aligned}$$

Therefore, the average power dissipation caused by the dc component is $\boxed{10\% \text{ of } P}$.